Statisitcs and probability

Descriptive statistics

Data collection

Qualitative data collection

1. Measure of central tendency
    1. MEAN
    2. MEDIAN
    3. MODE
    4. STANDARD DEVAITON
    5. VARIANCE
    6. CO-VARIANCE
    7. COEFFIECNT

FREQURENCY : it is the mode it shows the number of occurrences of a number in particular data

For ex: 1,3,4,5,67

Tell me the frequency of 3

Frequency of 3 is 1 because it only appers 1 time in the time

Grouped data : the data which is group in certain range

Age	marks
0-10	100
11-20	200
21-30	300

Grouped data is well organized and generally don't require any order

Ungrouped data/discrete

11,24,56,7,8,90,

Organise , clean if the raw data is there

Grouped data is also called as continuous series data

0-10,11-20-21-30(continuous series / class intervals)

Measures of central tendency on population

Mean – average

X= population

x- - sample mean

mean=sum(x)/n

there are the marks of 5 students find the mean

12,34,56,78,90

X= 12+34+56+78+90/5=54

54 is mean

Out of 100 students we have only the marks of 40 students who passed calculate the average marks obtained by passed students

10,20,40……n

x-= marks(sum)/40

median

odd

even

12 ,12,34,56,78

N=5

N is odd

Median = n+1/2

=5+1/2

=6/2

=3

34 is median

12 ,12,34,56,78,56,78

N=7

N is odd

Median = n+1/2

=7+1/2

=8/2

=4

56 is median

Even = (n/2)th observation +(n/2+1)th observation/2

((n/2)+(n/2+1))/2

N/2=6/2=3

Find the median of 2,4,6,8,10,12

Number of observation =6

Find the two middle positions

n/2=6/2=3

the middle obervations are 3 and 4 the 6 and 8

median =6+8/2

=14/2=7

Median will come between 6 and 8

Median is the measure of central tendency it never ever changes by average

Mode – it is the max number of occurences

12,12,34,56

Mode is 12